

Windows Communication Foundation (WCF)

Outline

- Introduction
- WCF Contracts
 - Service
 - Data
 - Message
- Bindings
- Security
- Reliability
- Declarative
- Summary

Introduction

- WCF is a unified programming model for building service oriented application.
- It is used to build
 - Secure
 - Reliable
 - Transacted
- applications that integrated across platforms and interoperate with existing applications.

Features of WCF

- Service orientation
- Loosely coupled
- Interoperability
- Multiple message patterns
 - Request-reply, one-way, duplex
- Publishing service metadata
 - WSDL, XSD, WS-Policy
- Data Contract
 - Classes

Features of WCF

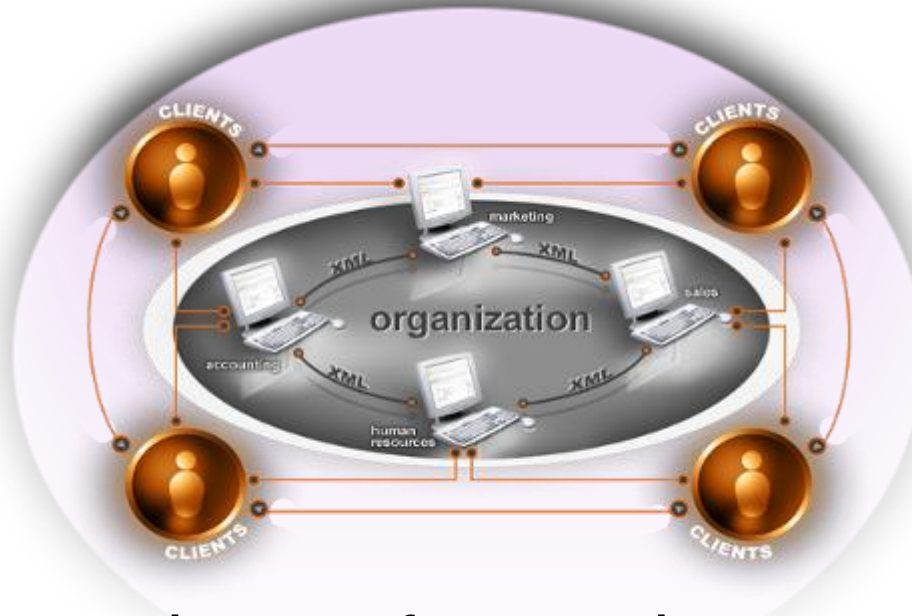
- Security
 - SSL, WS-SecureConversation
- Multiple Transports and Encodings
 - HTTP, TCP, named Pipe, MSMQ
 - Text, binary encoding (MTOM-Message Transmission Optimization Mechanism)
- Reliable and Queued Messages
 - WS-Reliable messaging and MSMQ
- Durable Messages
 - Message never lost during disruption in communication (saved to a database). Used in Windows Workflow foundation

Features of WCF

- Transactions
 - WS-AtomicTransaction, API, MS Distributed Transaction Coordinator
- AJAX and REST support
 - XML, ATOM, JSON
- Extensibility
- Multiple service hosting options
 - IIS, app-hosting

The Challenge

Radically Simplifying Distributed Application Development

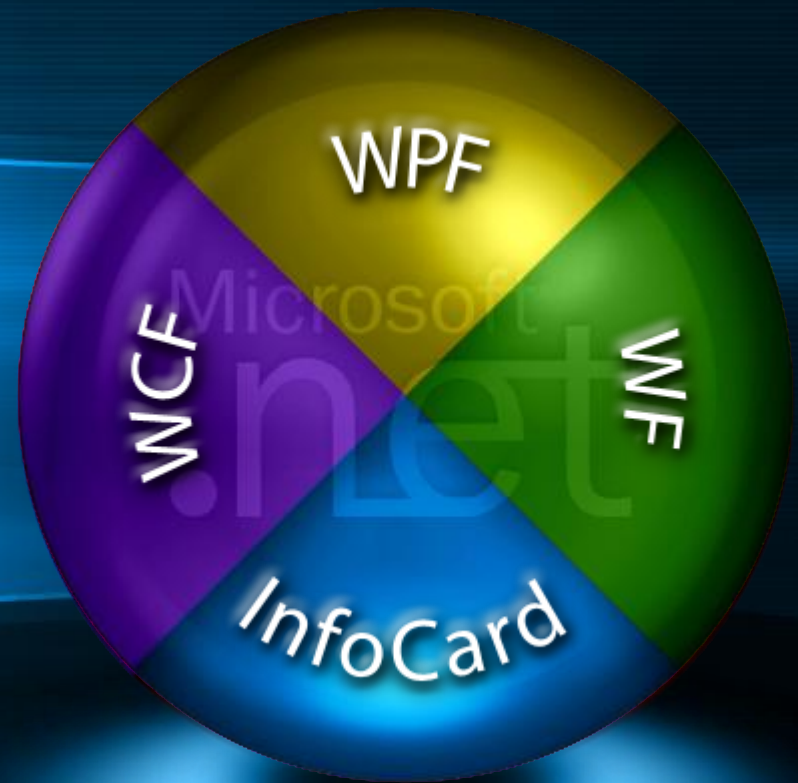


**Development of connected systems
remains costly and frustrating**

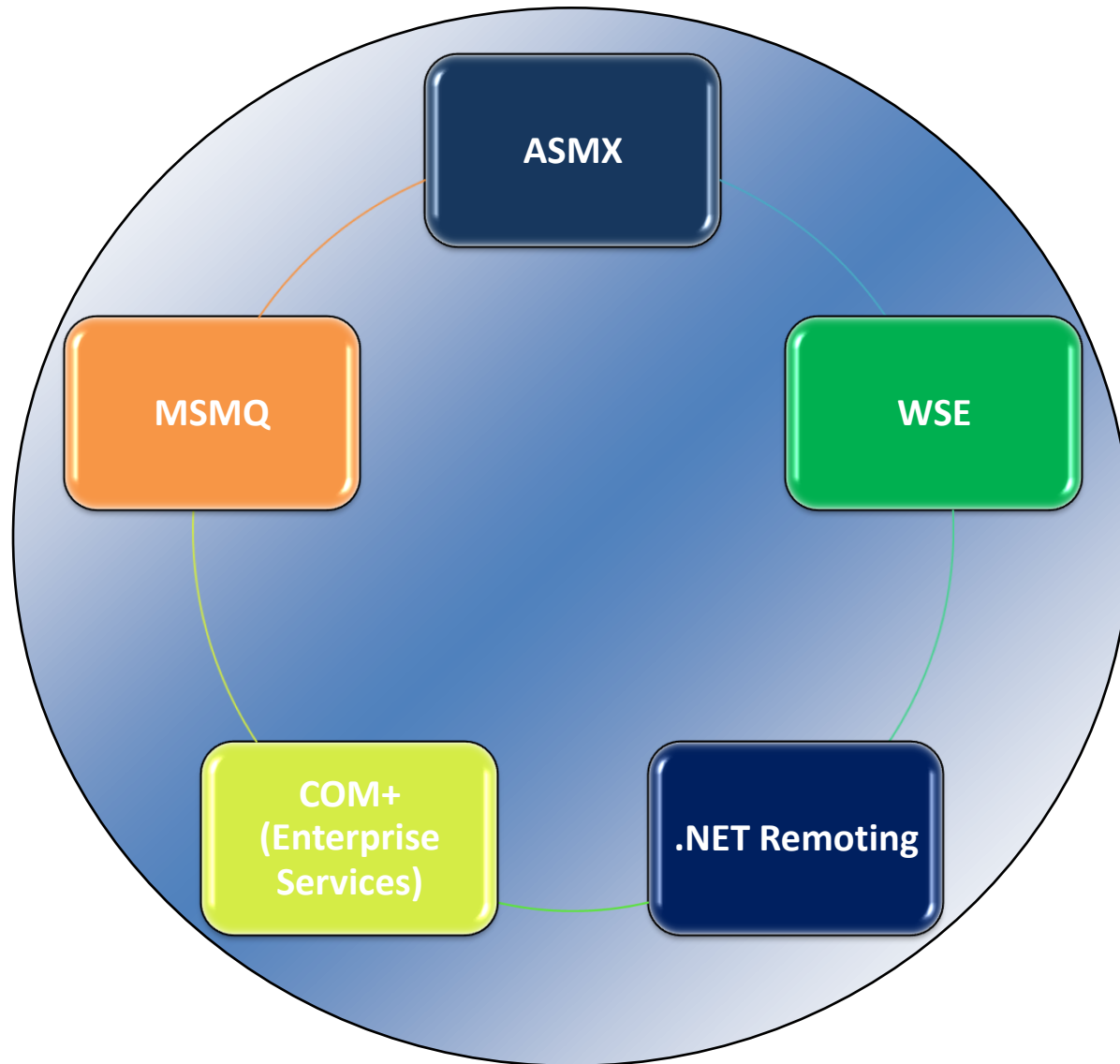
- Different programming models for different tasks
- Need for security and reliable messaging
- Interoperability with applications on other platforms
- Productive service-oriented programming model needed

Windows Communication Foundation

**Unified framework for
rapidly building
service-oriented applications**

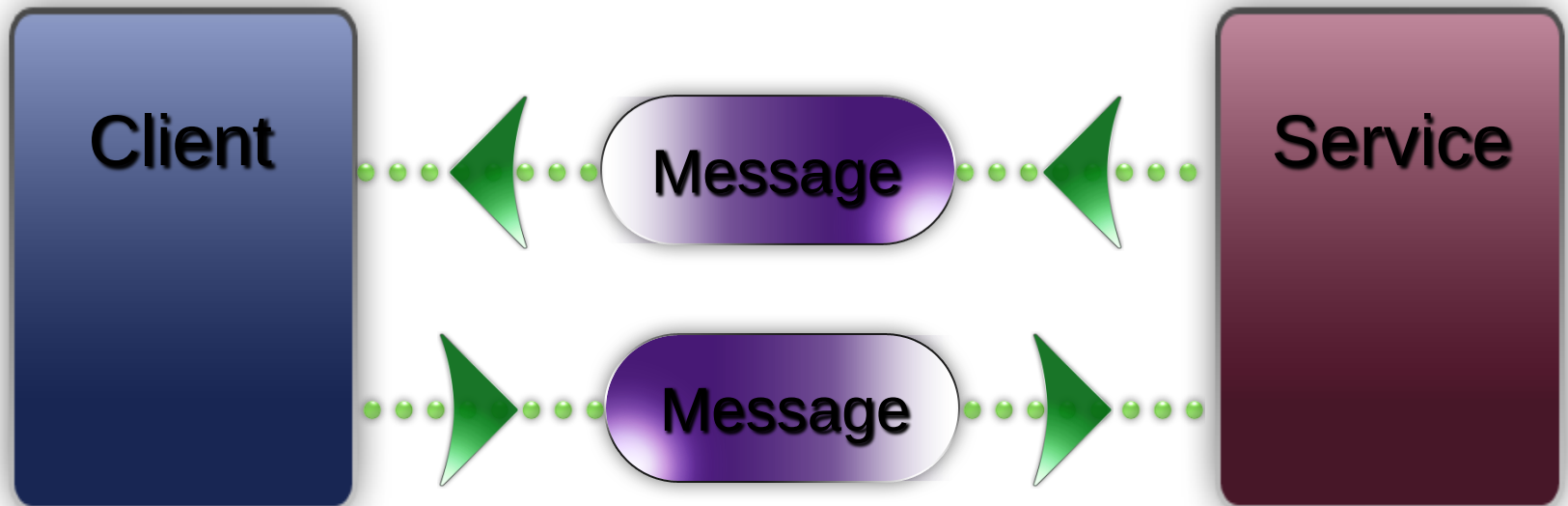


What Does WCF Replace?

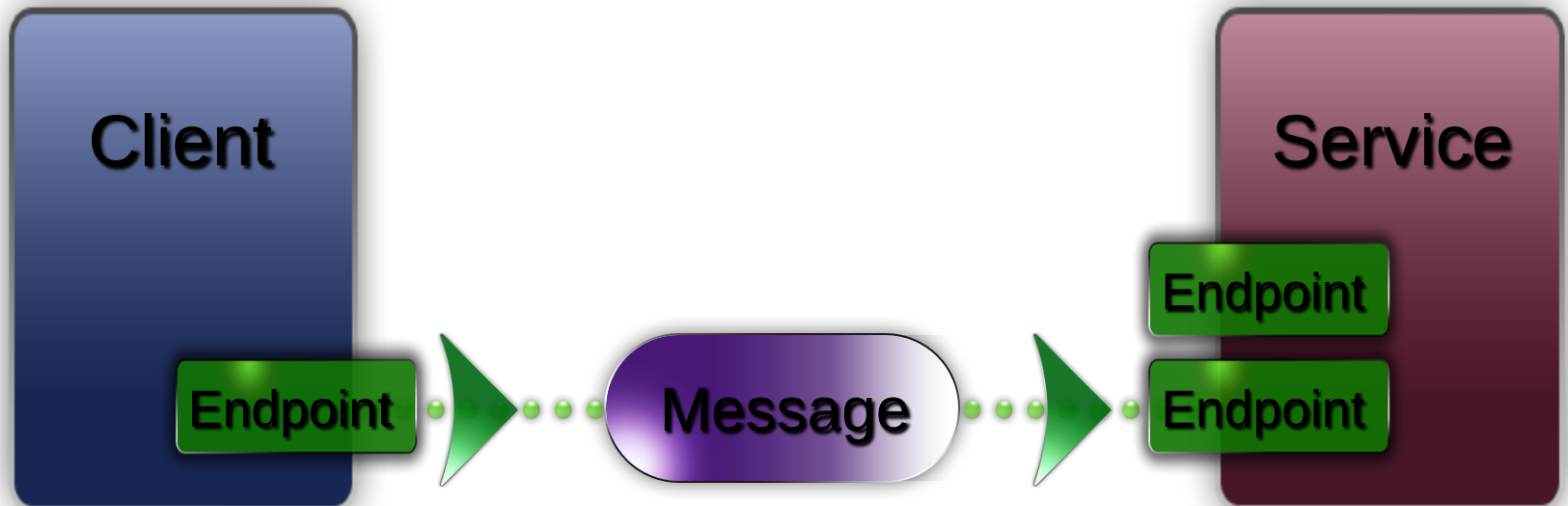


UNDERSTANDING WCF PRINCIPLES

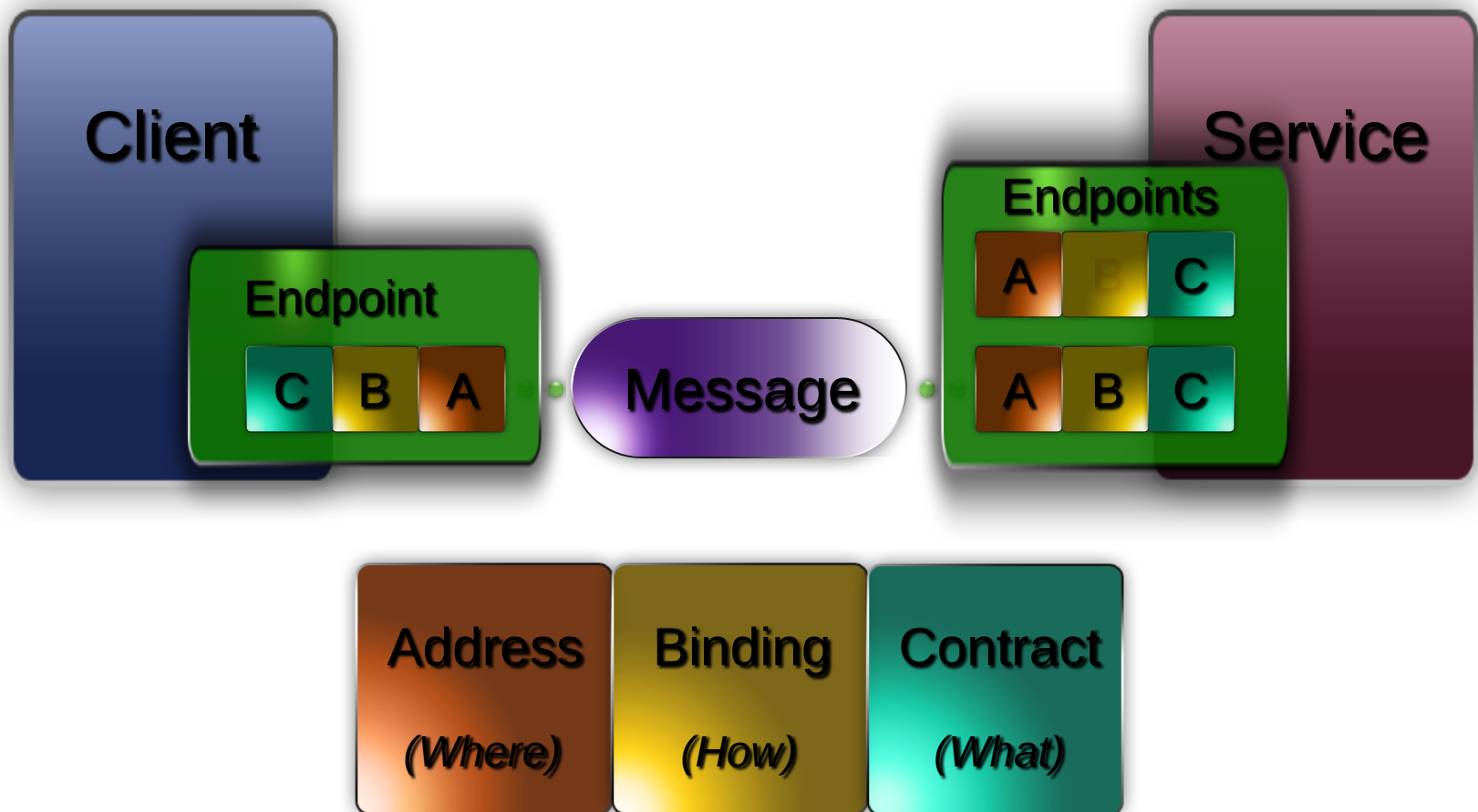
Services and Clients



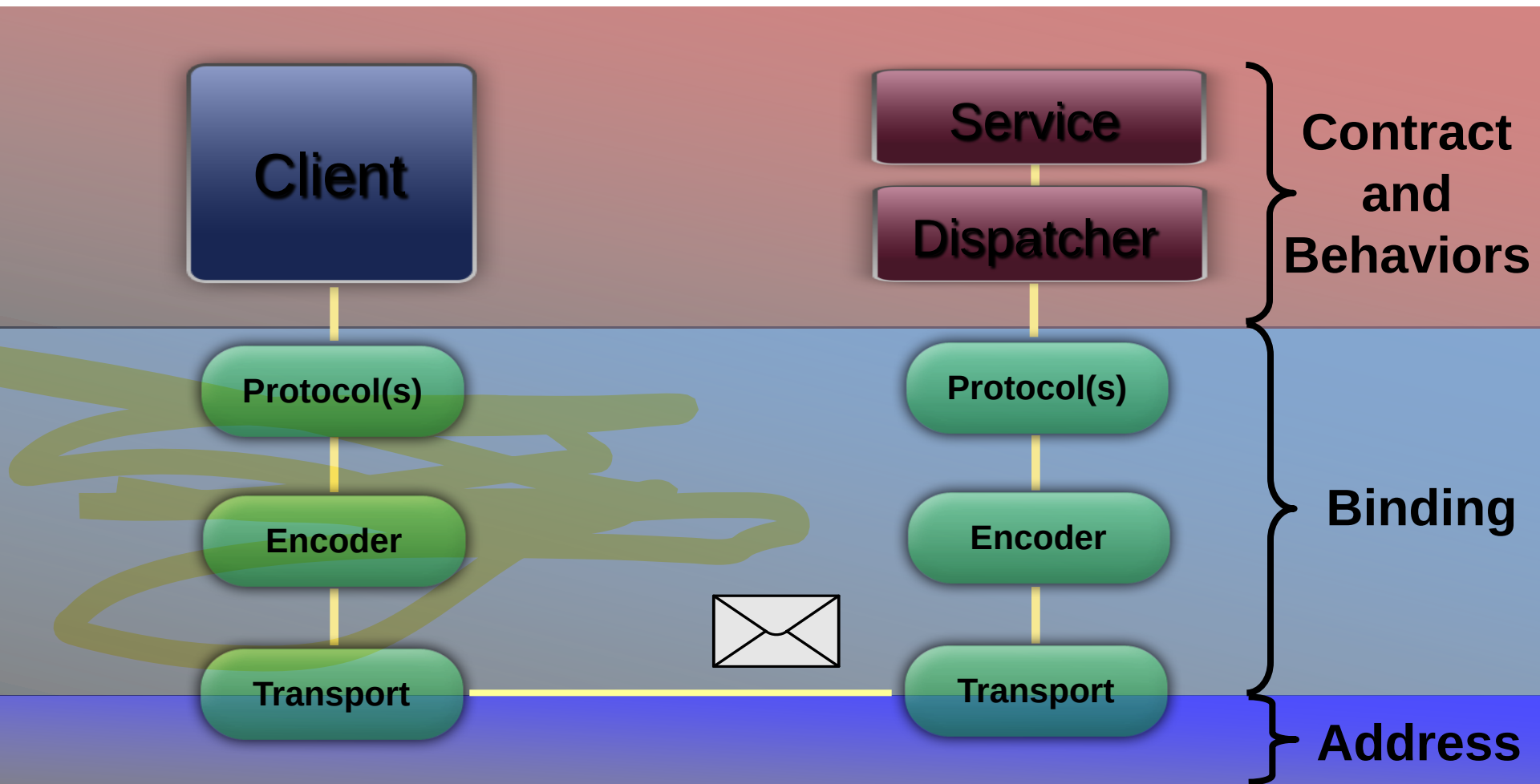
Endpoints



Address, Binding, Contract



WCF Architecture: Messaging Runtime



The what

CONTRACTS

Three Types of Contracts

Service Contract

Defines Operations,
Behaviors and
Communication
Shape

What does your
service do

Data Contract

Defines Schema and
Versioning Strategies

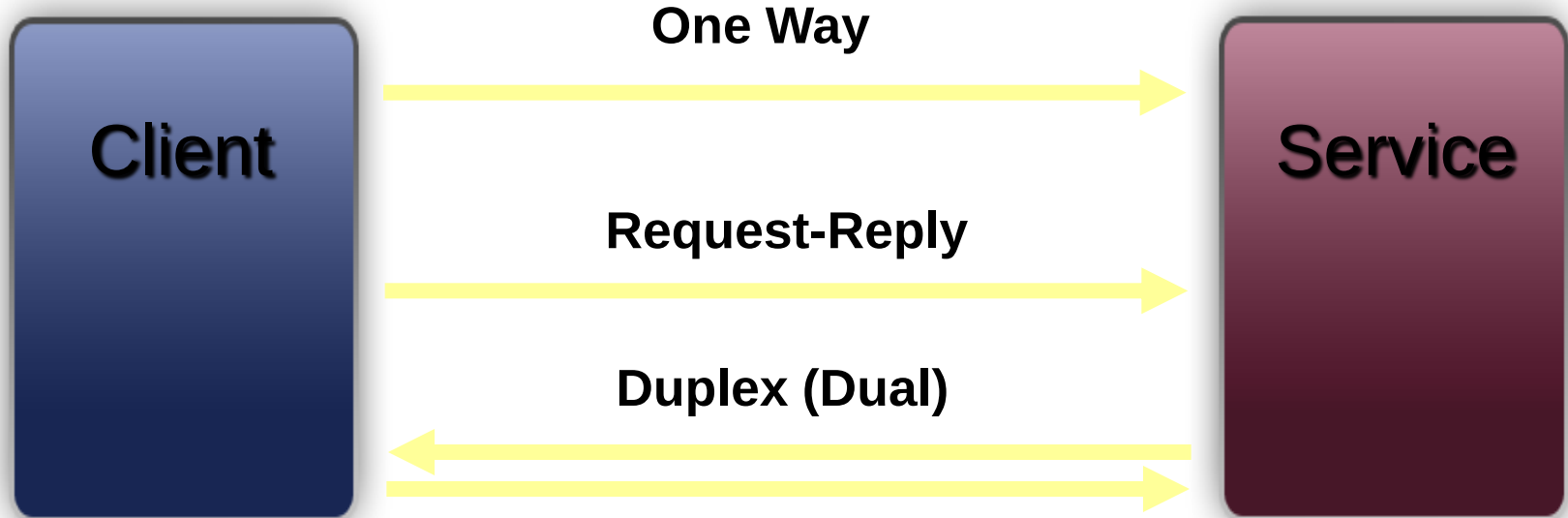
What object data is
used

Message Contract

Allows defining
application-specific
headers and
unwrapped body
content

Allows control over
the SOAP structure of
messages

Ways to Talk



- One Way:
 - Datagram-style delivery
- Request-Reply
 - Immediate Reply on same logical thread
- Duplex
 - Reply “later” and on backchannel (callback-style)

What does your service do?

SERVICE CONTRACTS

Service Contract

```
using System.ServiceModel;

[ServiceContract]
public interface ICalculate
{
    [OperationContract]
    double Add( double a, double b);
    [OperationContract]
    double Subtract( double a, double b);
}
```

Service Contract: OneWay

```
[ServiceContract]
public interface IOneWayCalculator
{
    [OperationContract(IsOneWay=true)]
    void StoreProblem (ComplexProblem p);
}
```

Service Contract: Duplex Asymmetric

```
[ServiceContract(Session=true,  
    CallbackContract=typeof(ICalculatorResults))]  
public interface ICalculatorProblems  
{  
    [OperationContract(IsOneWay=true)]  
    void SolveProblem (ComplexProblem p);  
}  
  
public interface ICalculatorResults  
{  
    [OperationContract(IsOneWay=true)]  
    void Results(ComplexProblem p);  
}
```

What object data needs to flow back and forth?

DATA CONTRACTS

Data Contract

[DataContract]

```
public class ComplexNumber  
{
```

[DataMember]

```
public double Real = 0.0D;
```

[DataMember]

```
public double Imaginary = 0.0D;
```

```
public ComplexNumber(double r, double i)
```

```
{
```

```
    this.Real = r;
```

```
    this.Imaginary = i;
```

```
}
```

```
}
```

Defines the mapping between the type and a SOAP envelope

MESSAGE CONTRACTS

Message Contract

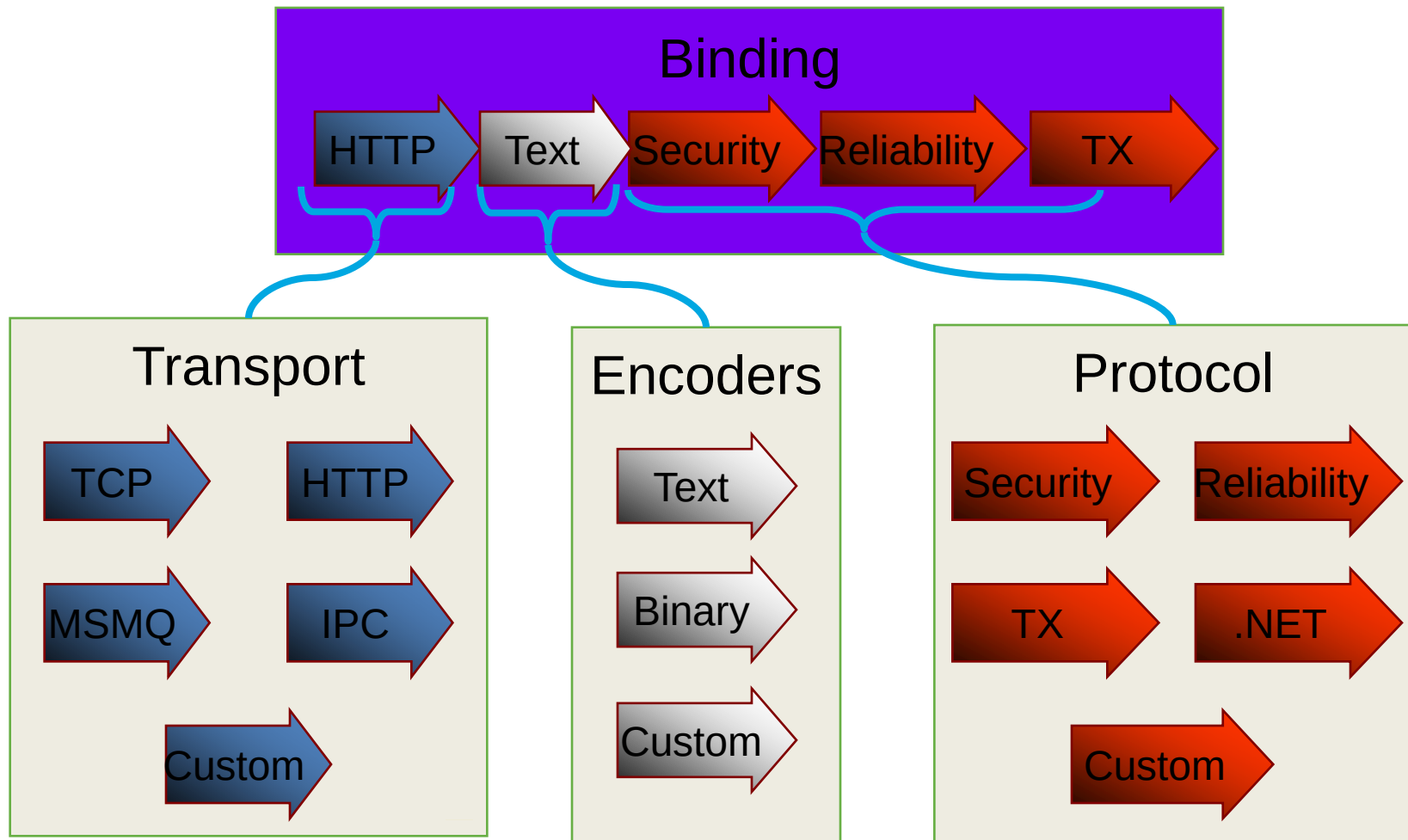
```
[MessageContract]
public class ComplexProblem
{
    [MessageHeader]
    public string operation;
    [MessageBody]
    public ComplexNumber n1;
    [MessageBody]
    public ComplexNumber n2;
    [MessageBody]
    public ComplexNumber solution;

    // Constructors...
}
```

How?

BINDINGS

Bindings & Binding Elements

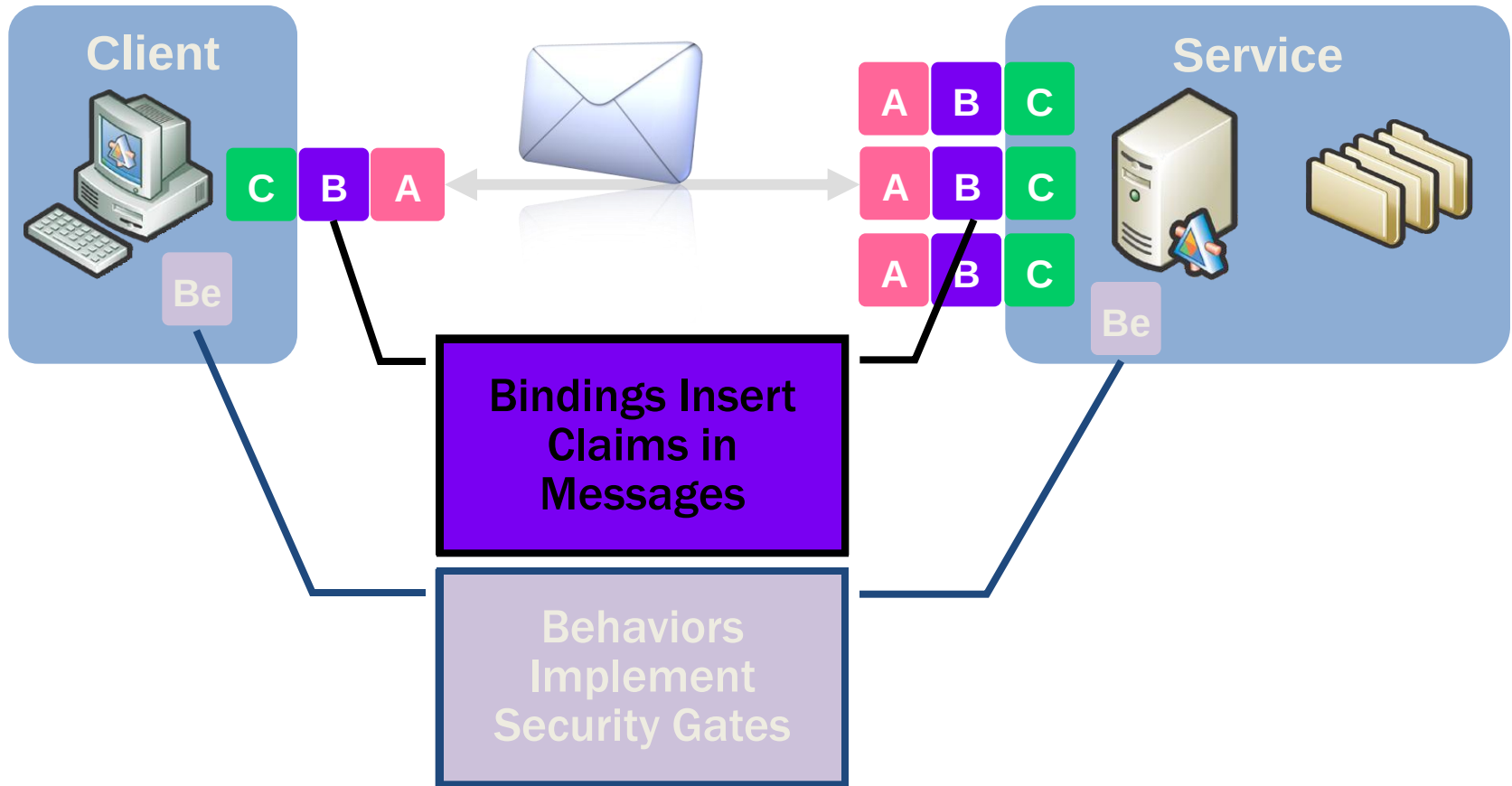


Standard Bindings

Binding	Interop	Security	Session	TX	Duplex
BasicHttpBinding	BP 1.1	N, T	N	N	n/a
WSHttpBinding	WS	<u>M</u>, T, X	<u>N</u>, T, RS	<u>N</u>, Yes	n/a
WSDualHttpBinding	WS	<u>M</u>	<u>RS</u>	<u>N</u>, Yes	Yes
WSFederationBinding	Federation	<u>M</u>	<u>N</u>, RS	<u>N</u>, Yes	No
NetTcpBinding	.NET	<u>T</u>, M	<u>T</u>, RS	<u>N</u>, Yes	Yes
NetNamedPipeBinding	.NET	<u>T</u>	<u>T</u>, N	<u>N</u>, Yes	Yes
NetPeerTcpBinding	Peer	<u>T</u>	<u>N</u>	<u>N</u>	Yes
NetMsmqBinding	.NET	<u>T</u>, M, X	<u>N</u>	<u>N</u>, Yes	No
MsmqIntegrationBinding	MSMQ	<u>T</u>	<u>N</u>	<u>N</u>, Yes	n/a

N = None | T = Transport | M = Message | B = Both | RS = Reliable Sessions

Bindings & Behaviors: Security

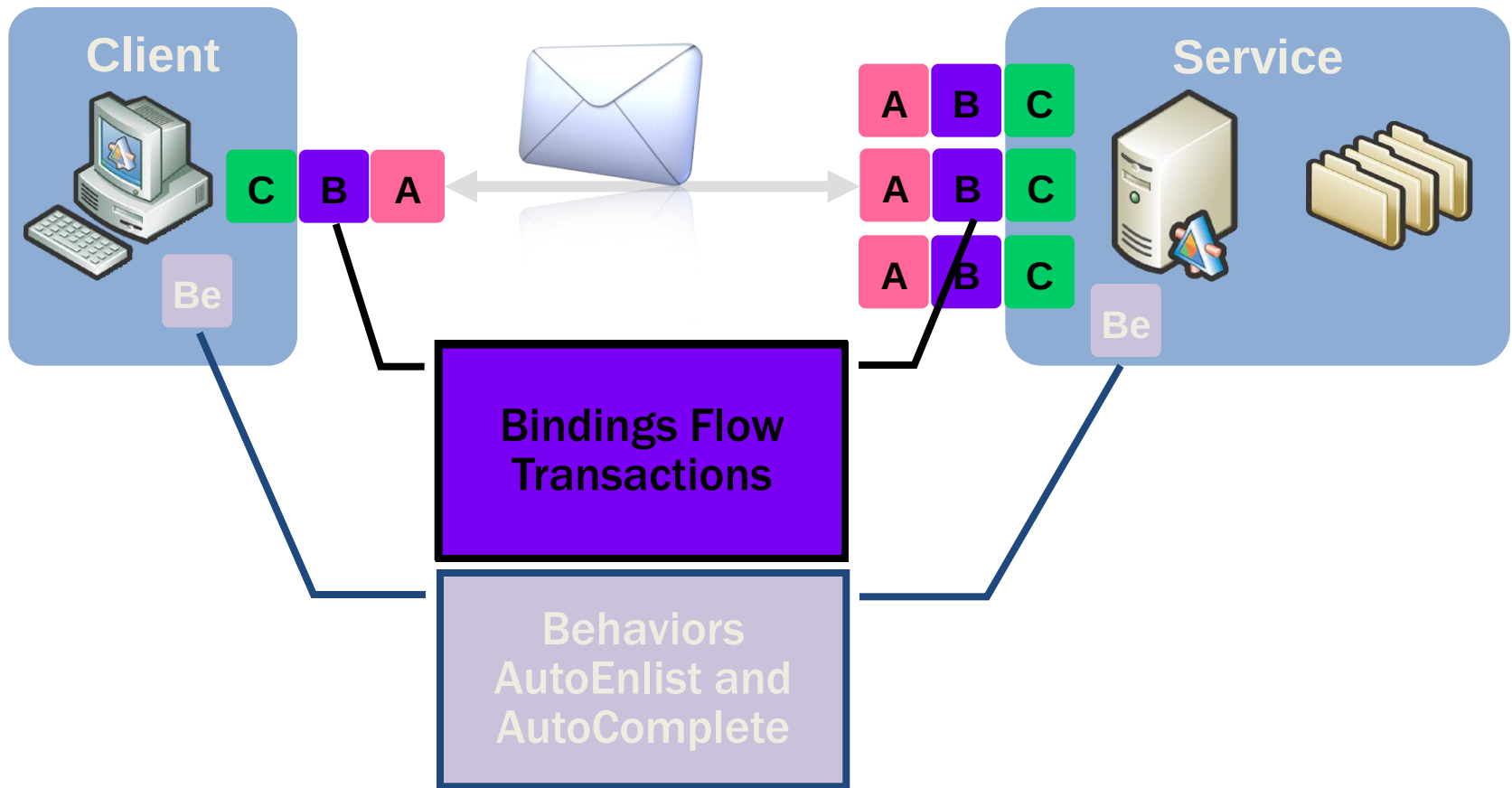


Feature Overview

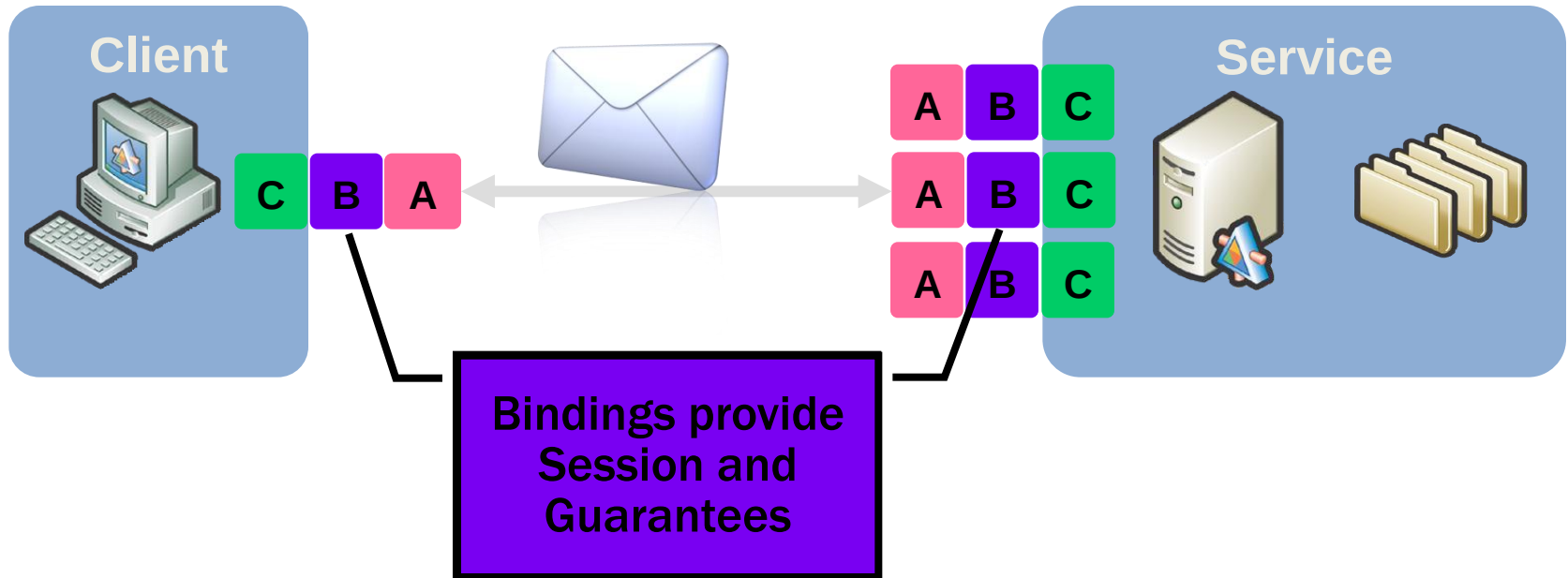
Security

- Claims based end-to-end security
 - Secure end-to-end message exchanges
 - Secure access to resources
 - Record resource access requests
- X509, Username/Password, Kerberos, SAML, custom credentials
- Message security
 - Confidentiality and integrity
 - Transport or message level
- Access to resources
 - Authentication and authorization

Bindings & Behaviors: Transactions



Bindings & Behaviors: Reliable Sessions



Feature Overview

Reliability and Transactions

- End-to-end Reliable messaging
 - In-order guarantees
 - Exactly once guarantees
- Transport-Independent Sessions
 - Integration with ASP.NET Sessions in IIS-Hosted compatibility mode
- Transactions
 - Guaranteed atomic success or failure across services

Code vs. Config

Defining Endpoints

```
<?xml version="1.0" encoding="utf-8" ?>
<configuration xmlns="http://schemas.microsoft.com/.NetConfiguration/v2.0">
  <system.serviceModel>
    <services>
      <service serviceType="CalculatorService">
        <endpoint address="Calculator"
          bindingSectionName="basicProfileBinding"
          contractType="ICalculator" />
      </service>
    </services>
  </system.serviceModel>
</configuration>
```

Configuring Bindings

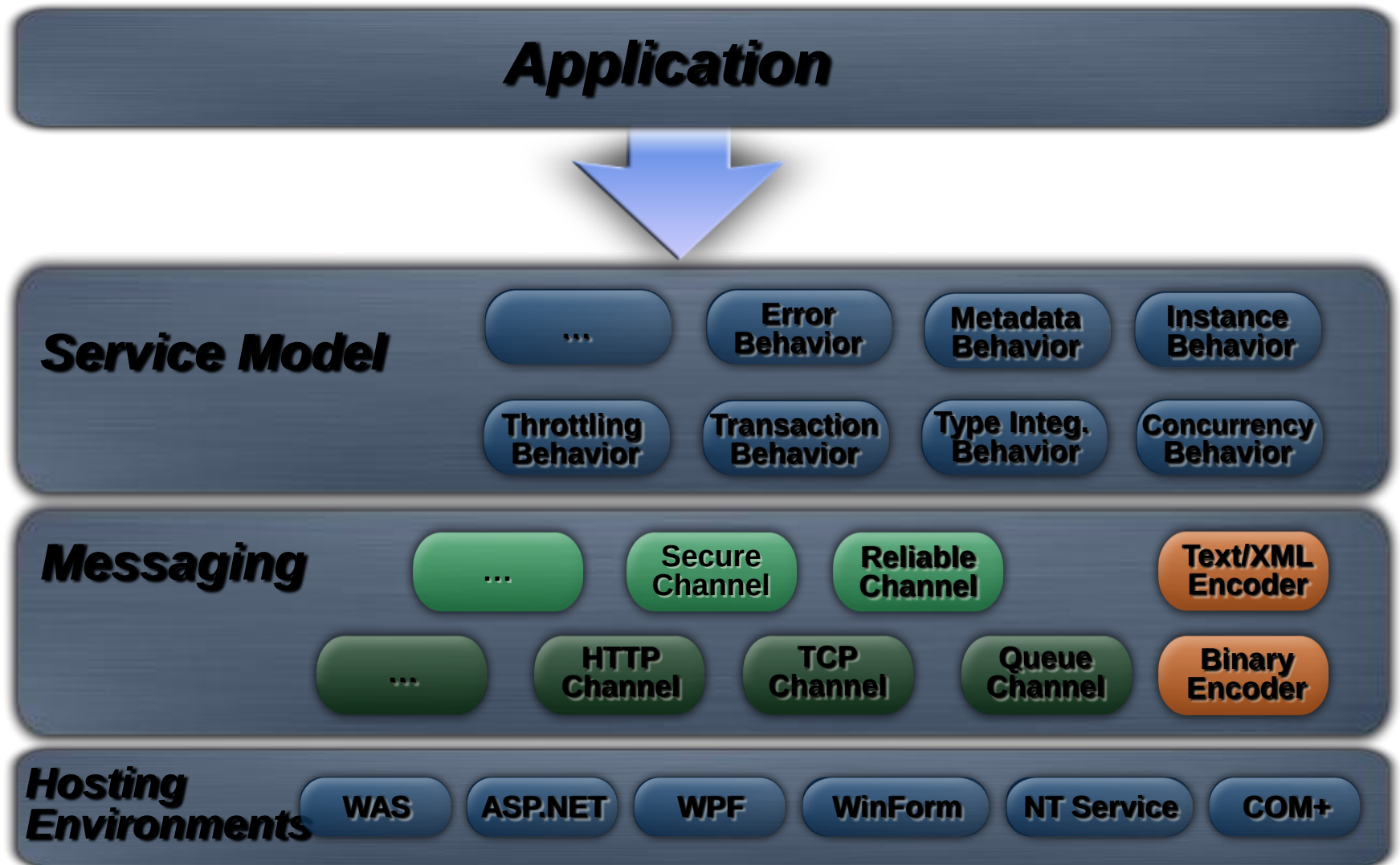
```
<endpoint address="Calculator"
  bindingSectionName="basicProfileBinding"
  bindingConfiguration="Binding1"
  contractType="ICalculator" />
```

```
<bindings>
  <basicProfileBinding>
    <binding configurationName="Binding1"
      hostnameComparisonMode="StrongWildcard"
      transferTimeout="00:10:00"
      maxMessageSize="65536"
      messageEncoding="Text"
      textEncoding="utf-8"
    </binding>
  </basicProfileBinding>
</bindings>
```

Custom Bindings

```
<bindings>
  <customBinding>
    <binding configurationName="Binding1">
      <reliableSession bufferedMessagesQuota="32"
        inactivityTimeout="00:10:00"
        maxRetryCount="8"
        ordered="true" />
      <httpsTransport manualAddressing="false"
        maxMessageSize="65536"
        hostnameComparisonMode="StrongWildcard"/>
      <textMessageEncoding maxReadPoolSize="64"
        maxWritePoolSize="16"
        messageVersion="Default"
        encoding="utf-8" />
    </binding>
  </customBinding>
</bindings>
```

WCF Summary



WCF Summary

- WCF is the future of distributed computing
- It combines the best of all existing Microsoft distributed computing stacks
- It uses WS-* standards for interoperability and .NET value-add for performance and integration with existing solutions
- WCF is available for Windows Vista, Windows XP SP2, Windows Server 2003, Windows Server 2008